

Macroeconomia I

Pedro Garcia Duarte

INSPER

Mestrado Profissional em Economia



What this class is about

- Macroeconomics: looking at the economy as a whole
- Two main questions:
 - ▶ Why is there economic growth?
 - ▶ Why are there business cycles?
- Objects we measure and study
 - ▶ GDP
 - ▶ consumption
 - ▶ saving
 - ▶ investment
 - ▶ balance of payments
 - ▶ employment
 - ▶ unemployment
 - ▶ inflation
 - ▶ interest rates



Approach

Abstract mathematical models
+
Contemporary and historical applications

- Emphasis on relating abstract theory to applications

A set of small, light blue navigation icons typically found in Beamer presentations, including symbols for back, forward, search, and other navigation functions.

Materials

- Pablo Kurlat (2020) *A Course in Modern Macroeconomics*
<https://sites.google.com/view/pkurlat/a-course-in-modern-macroeconomics>
- Optional textbooks (not recommended)
 - ① Wendy Carlin and David Soskice (2006) *Macroeconomics: Imperfections, Institutions and Policies*
 - ② Charles I. Jones (2020) *Macroeconomics*
 - ③ David Romer (2018) *Advanced Macroeconomics*
 - ④ Stephen D. Williamson (2020) *Macroeconomics*

A set of small, light blue navigation icons typically found in Beamer presentations, including symbols for back, forward, search, and other navigation functions.

Grading

- 7 problem sets
 - ▶ Each will contain not-so-hard and difficult questions
 - ▶ Count the best 5
- Final Exam (September 28)

Problem Sets	35%
Final	65%

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ≡ ↺ 🔍 ↻

Topics

- 1 Measurement and National Accounting
- 2 Economic Growth
- 3 Microeconomic Foundations
 - ▶ Consumption and Savings
 - ▶ Labor and Leisure
 - ▶ General Equilibrium Theory
- 4 Money and Inflation
- 5 New Keynesian AD-AS model for policy evaluation

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ≡ ↺ 🔍 ↻

Measurement

- What we measure and how
 - ▶ How it differs from what we'd like to measure
- Main macroeconomic variables
 - ▶ GDP accounting and its shortcomings
 - ▶ Price indices
 - ▶ Labor market indicators
- Alternatives to GDP
 - ▶ Measures of welfare based on economic theory



Economic Growth

- Evidence on economic growth across countries and over time
- The Solow model of economic growth
 - ▶ The role of technological progress and capital accumulation
- Contrasting the model and evidence
 - ▶ Successes, failures and open questions



Microeconomic Foundations

- Key decisions in the Solow model are modeled as exogenous
 - ▶ Labor supply
 - ▶ Savings rate
- What incentives govern these decisions?
 - ▶ Consumption - savings problems
 - ▶ Labor - leisure problems
- General equilibrium
 - ▶ How individual decisions add up to aggregate outcomes
 - ▶ The First Welfare Theorem

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ≡ ▶ ↺ 🔍 ↻

Money, Inflation and Policy Evaluation

- What is money?
- The demand for money
- Equilibrium in the money market
- Inflation: revenues and costs
- AD-AS New Keynesian model (microfoundations) for policy evaluation

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ≡ ▶ ↺ 🔍 ↻

Measuring GDP and Living Standards: GDP

Pedro Garcia Duarte

INSPER

Mestrado Profissional em Economia



Outline

- 1 GDP Accounting
- 2 Nominal and Real GDP
- 3 Comparing Across Countries and PPP



GDP

- A measure of total production in a country in a year

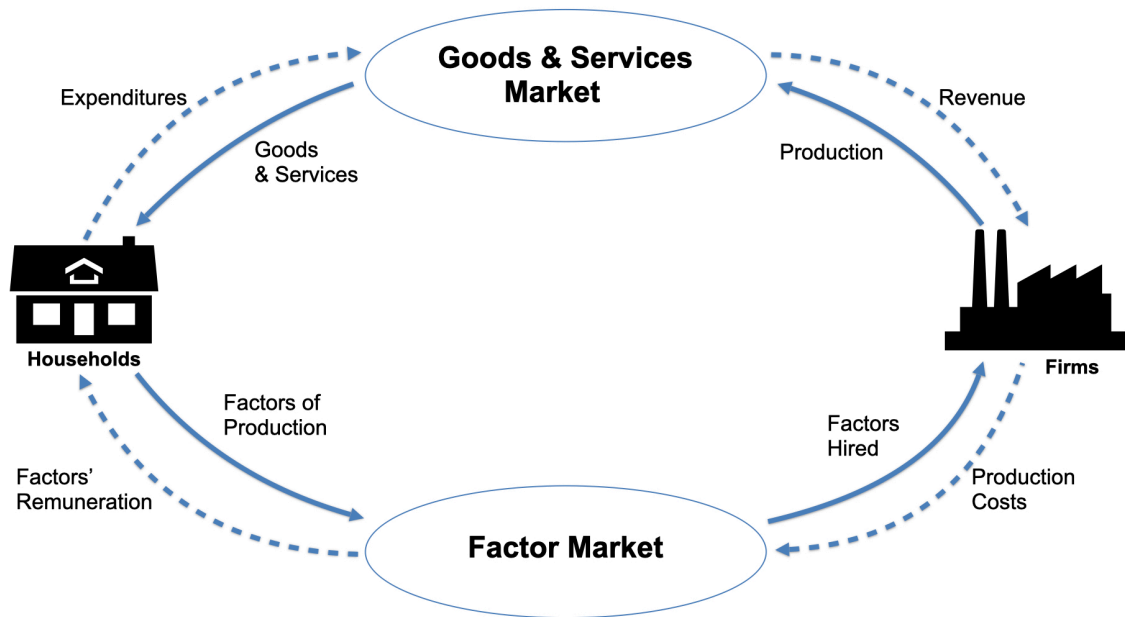


Figure: Circular Flow of Income in a Simplified Economy

Navigation icons: back, forward, search, etc.

GDP

- Three ways to look at National Product: Production, Expenditure, and Income
- Three equivalent methods for adding up:

Production	Income	Expenditure
Agriculture	Employee compensation	Consumption
Manufacturing	Corporate profits	Investment
Services	Proprietor's income	Government
Government	Rental Income	Net Exports
etc.	Interest Income	
	Depreciation	

- Different levels of details possible
- Expenditure method, in a formula

$$Y = C + I + G + X - M$$

Navigation icons: back, forward, search, etc.

GDP in the US - 2023

Production		Income		Expenditure	
Agriculture	251	Employee Comp.	14,234	Consumption	18,570
Mining	381	Corporate Profits	3,258	Investment	4,844
Utilities	434	Proprietor's income	1,849	Government	4,745
Construction	1204	Rental income	967	Exports	3,027
Manufacturing	2,805	Bus.Trsf. Payments	215	Imports	3,826
Wholesale + Retail	3,352	Interest Income	186		
Transport	970	Taxes	1,732		
Media	1,475	Surpl. Gov. Enterpr.	-10		
Finance + Insurance	1,988	Depreciation	4,586		
Real Estate	3,668	Net Inc. to RoW	-164		
Professional services	3,544	Statist. discrepancy	507		
Education + Health	2,352				
Arts + Entertainmt.	1,231				
Other services	597				
Government	3,108				
Total	27,360		27,360		27,360

Table: Figures in billions of dollars. Source: BEA.

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ↺ 🔍 ↻

Basic Example

Example

- Amy, who is self-employed, produces lettuce in her garden
- She sells it to Bob for \$1.
- Bob eats it.

Production		Income		Expenditure	
Agriculture	1	Proprietor's income	1	Consumption	1

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ↺ 🔍 ↻

The Principle of Value Added

- Production is typically multi-stage: Value Added
- We want the value at the end of the process, avoiding double counting

Example

- Amy owns Fertilizer, Inc.
- Fertilizer, Inc. hires Bob and pays him a wage of \$0.50.
- Fertilizer, Inc. sells the fertilizer to Carol, a self-employed farmer, for \$0.80.
- Carol uses it to produce lettuce, which she sells to Daniel for \$1.
- Daniel eats the lettuce.

Production		Income		Expenditure	
Manufacturing	0.8	Wages	0.5	Consumption	1
Agriculture	0.2	Profits	0.3		
		Proprietor's income	0.2		
Total	1.0	Total	1.0	Total	1.0

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ↺ 🔍 ↻

Different Forms of Investment

- Investment = producing something to be used for production in the future (including inventories and houses)

Example

- General Electric builds an X-ray machine.
- GE sells the machine to Stanford Hospital for \$1,000.
- The cost of producing it is made up of workers' wages of \$600.

Production		Income		Expenditure	
Manufacturing	1,000	Wages	600	Investment	1,000
		Corporate Profits	400		
Total	1,000	Total	1,000	Total	1,000

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ↺ 🔍 ↻

Different Forms of Investment

- Investment = producing something to be used for production in the future (including inventories and houses)

Example

- Dunder Mifflin produces 500 tons of white paper worth \$40,000
- It sells 200 tons and stores the remaining in its warehouse while it waits for customers to buy them.
- The cost of producing them is made up of workers' wages of \$50,000.

Production		Income		Expenditure	
Manufacturing	40,000	Wages	50,000	Investment	24,000
		Corporate Profits	-10,000	Consumption	16,000
Total	40,000	Total	40,000	Total	40,000

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ↺ 🔍 ↻

Durables

- Distinction between Investment and Consumption is at times not so clear
- Residential construction is investment
- Cars, refrigerators, clothes, etc. are not (Durable Consumption)

Example

- A property developer builds a house (at zero cost) and sells it to Claire for \$100,000.

Production		Income		Expenditure	
Construction	100,000	Corporate Profits	100,000	Investment	100,000

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ↺ 🔍 ↻

The Government

- Gov't goods and services are valued at production costs
- "G" includes both gov. consumption and investment

Example

- The City of Palo Alto hires the Los Angeles Philharmonic to play a free concert in Stanford Stadium.
- The musicians are paid \$65,000 and renting the stadium costs \$20,000.

Production		Income		Expenditure	
Public Concert	85,000	Wages Rental Income	65,000 20,000	Government	85,000
Total	85,000	Total	85,000	Total	85,000



Depreciation

- GDP ("Gross") = output before depreciation

Example

- Zak's Transport Co. owns a fleet of taxis.
- They are all new at the beginning of the year, worth a total of \$1,000.
- A taxi depreciates completely in 5 years.
- During the course of the year, the company pays its workers \$200 in wages, has no other costs, and collects \$500 in fares.

Production		Income		Expenditure	
Transp. Service	500	Corporate Profits:		Consumption	500
		Revenue	500		
		Wages	-200		
		Depreciation	-200		
		Total Profit	100		
		Wages	200		
		Depreciation	200		
Total	500	Total	500	Total	500



Outline

- 1 GDP Accounting
- 2 Nominal and Real GDP
- 3 Comparing Across Countries and PPP

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ↺ 🔍 ↻

Nominal and Real GDP

- **Nominal GDP** = total value of goods and services produced, valued at prices they had at the time they were produced
- **Real GDP** = total value of goods and services produced, valued at constant prices (thus, comparable)

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ↺ 🔍 ↻

Changing *Relative* Prices

Example

2014				2015			
Good	Qty	Unit Price	Total	Good	Qty	Unit Price	Total
Haircuts	10	50	500	Haircuts	11	60	660
Cars	1	1,000	1,000	Cars	2	600	1,200
			1,500				1,860

- **Option 1** - Base year prices: $Y_t = \sum_i p_{i0} \times q_{it}$

2015			
Good	Qty	Base Year Price	Total
Haircuts	11	50	550
Cars	2	1,000	2,000
			2,550

- Growth rate:

$$g = \frac{2,550}{1,500} - 1 = 70\%$$



Changing *Relative* Prices

Example

2014				2015			
Good	Qty	Unit Price	Total	Good	Qty	Unit Price	Total
Haircuts	10	50	500	Haircuts	11	60	660
Cars	1	1,000	1,000	Cars	2	600	1,200
			1,500				1,860

- **Option 2** - Final year prices: $Y_0 = \sum_i p_{it} \times q_{i0}$

2014			
Good	Qty	Final Year Price	Total
Haircuts	10	60	600
Cars	1	600	600
			1,200

- Growth rate:

$$g = \frac{1,860}{1,200} - 1 = 55\%$$



Changing *Relative* Prices

Example

2014				2015			
Good	Qty	Unit Price	Total	Good	Qty	Unit Price	Total
Haircuts	10	50	500	Haircuts	11	60	660
Cars	1	1,000	1,000	Cars	2	600	1,200
			1,500				1,860

- **Option 3** - Chained prices:

$$g = (1 + g^I)^{0.5} (1 + g^F)^{0.5} - 1 = 0.7^{0.5} \times 0.55^{0.5} \approx 0.62$$

- Real 2015 GDP at 2014 (chained) prices:

$$1,500 \times (1 + 0.62) \approx 2,435$$

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ≡ ≡ ≡ ≡ ≡ ≡ ≡ ≡

Outline

1 GDP Accounting

2 Nominal and Real GDP

3 Comparing Across Countries and PPP

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ≡ ≡ ≡ ≡ ≡ ≡ ≡ ≡

Converting Currencies

- 2018 data:

	United States	Mexico
GDP	20.5 trillion dollars	23.5 trillion pesos
Population	327 million	127 million
GDP per capita	62,700 dollars per person	185,000 pesos per person

- **Market** exchange rates

$$e = 0.053 \text{ dollars per peso}$$

- Mexican GDP per capita

$$185,000 \text{ pesos} \times 0.053 \text{ dollars per peso} = 9,805 \text{ dollars}$$

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ≡ ≡ ↺ 🔍 ↻

Purchasing Power Parity

- Problem: low (foreign) GDP in US\$
 - ▶ Output is actually lower
 - ▶ Prices are lower even *after* converting currencies

How to tell the difference?

- GDP at “purchasing power parity”: take Mexican goods and look for their equivalents in the US, take these US prices to value the Mexican goods in US dollars

$$\text{GDP in Foreign Country} = \sum_{i=1}^N p_i^{US} \times q_i$$

(in dollars, at PPP)

- By-product of PPP calculations: PPP exchange rate

$$\text{PPP exchange rate} \equiv \frac{\text{GDP in dollars at PPP}}{\text{GDP in foreign currency}}$$

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ≡ ≡ ↺ 🔍 ↻

Measuring GDP and Living Standards: Beyond GDP

Pedro Garcia Duarte

INSPER

Mestrado Profissional em Economia

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ≡ ≡ ↺ 🔍 ↻

Economics, Output and Wealth

The annual labour of every nation is the fund which originally supplies it with all the necessaries and conveniences of life which it annually consumes, and which consist always either in the immediate produce of that labour, or in what is purchased with that produce from other nations.

According therefore as this produce, or what is purchased with it, bears a greater or smaller proportion to the number of those who are to consume it, the nation will be better or worse supplied with all the necessaries and conveniences for which it has occasion.

(Adam Smith, 1776, *Wealth of Nations*)

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ≡ ≡ ↺ 🔍 ↻

Economics, Output and Wealth

Political Economy or Economics is a study of mankind in the ordinary business of life; it examines that part of individual and social action which is most closely connected with the attainment and with the use of the material requisites of wellbeing.

Thus it is on the one side a study of wealth; and on the other, and more important side, a part of the study of man. For man's character has been moulded by his every-day work, and the material resources which he thereby procures, more than by any other influence unless it be that of his religious ideals; and the two great forming agencies of the world's history have been the religious and the economic. (...)

And, in addition to the Residuum, there are vast numbers of people both in town and country who are brought up with insufficient food, clothing, and house-room; whose education is broken off early in order that they may go to work for wages; who thenceforth are engaged during long hours in exhausting toil with imperfectly nourished bodies, and have therefore no chance of developing their higher mental faculties.

(Alfred Marshall, 1890, *Principles of Economics*)

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ↺ 🔍 ↻

Economics, Output and Wealth

But even if we act to erase material poverty, there is another greater task, it is to confront the poverty of satisfaction - purpose and dignity - that afflicts us all. Too much and for too long, we seemed to have surrendered personal excellence and community values in the mere accumulation of material things. Our Gross National Product, now, is over \$800 billion dollars a year, but that Gross National Product - if we judge the United States of America by that - that Gross National Product counts air pollution and cigarette advertising, and ambulances to clear our highways of carnage. It counts special locks for our doors and the jails for the people who break them. It counts the destruction of the redwood and the loss of our natural wonder in chaotic sprawl. It counts napalm and counts nuclear warheads and armored cars for the police to fight the riots in our cities. It counts Whitman's rifle and Speck's knife, and the television programs which glorify violence in order to sell toys to our children.

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ↺ 🔍 ↻

The HDI

- UN's *Human Development Index* (HDI): an index between 0 and 1 with three components
 - ▶ Income
 - ▶ Education
 - ▶ Life Expectancy
- Indices:

$$\text{Life Expectancy Index} = \frac{\text{Life Expectancy} - 20}{85 - 20}$$

$$\text{Education Index} = \frac{1}{2} \left(\frac{\text{Avg. school yrs. 25-year-olds}}{15} + \frac{\text{Expected schools yrs. 5-year-olds}}{18} \right)$$

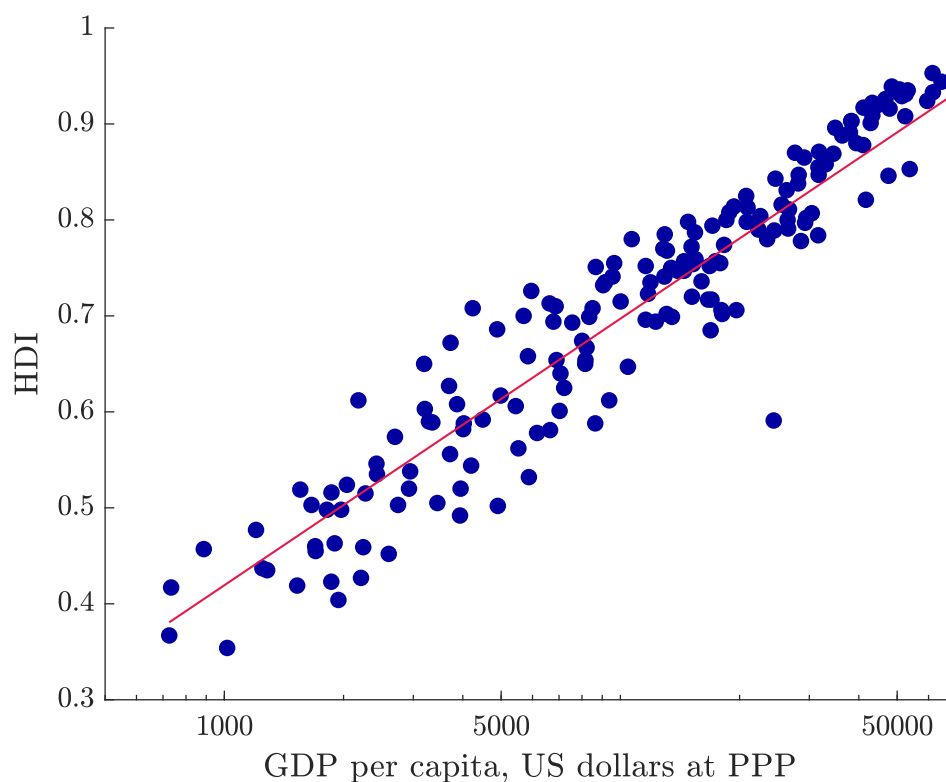
$$\text{Income Index} = \frac{\log(\text{GNP per capita}) - \log(100)}{\log(75,000) - \log(100)}$$

- HDI formula:

$$\text{HDI} = \sqrt[3]{\text{Life Expectancy Index} \times \text{Education Index} \times \text{Income Index}}$$

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ↺ 🔍 ↻

The HDI



◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ↺ 🔍 ↻

The HDI and Beyond

- HDI is unsatisfying for not being grounded on economic theory
 - ▶ What exactly is the HDI measuring?
 - ▶ Why convert variables into indices?
 - ▶ Why equal weights on the three factors?
- Jones and Klenow (2016): same spirit as HDI but with different methodology
- They focus on:
 - ▶ Consumption (instead of production)
 - ▶ Value of leisure and nonmarket production
 - ▶ Life expectancy
 - ▶ Inequality

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ≡ ↺ 🔍 ↻

Jones & Klenow's Methodology: **Living Standards?**

- “Rawls” spends a year as a resident in one of two countries
- Rawls doesn't know beforehand what individual he will be (young, old, rich, poor, ...)
- Possible two countries:
 - ▶ Utilia: country whose living standards we are trying to measure, with N inhabitants
 - ▶ A country like the US but with everyone's consumption multiplied by λ
- What value of λ makes Rawls indifferent between the two countries?
- This “equivalent variation” is the measure of living standards in Utilia

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ≡ ↺ 🔍 ↻

Jones & Klenow's Methodology: **Living Standards?**

- “Rawls” is utilitarian, caring about consumption, c (in fact, $C+G$), leisure, l , and life expectancy, a ($0 = \text{dead}$, $1 = \text{alive}$):

$$u(c, l, a) = \mathbb{E} \left[\left(\bar{u} + \frac{c^{1-\sigma}}{1-\sigma} - \theta(1-l)^2 \right) a \right]$$

- We find λ , given parameters, by solving:

$$u(\lambda c^{US}, l^{US}, a^{US}) = u(c^{Utilia}, l^{Utilia}, a^{Utilia})$$

- Consumption is what matters for Rawls, so it includes private and public consumption
- Rawls knows how much each Utilian consumes, but he doesn't know which one of them he will be

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ↺ 🔍 ↻

Jones & Klenow's Methodology: **Veil of Ignorance, Inequality and Risk Aversion**

- John Rawls (1921-2002): just decisions are those made behind the veil of ignorance
 - ▶ “Difference Principle:” society organized to benefit the least-well-off person

- Under the veil of ignorance “Rawls's” **expected utility** is:

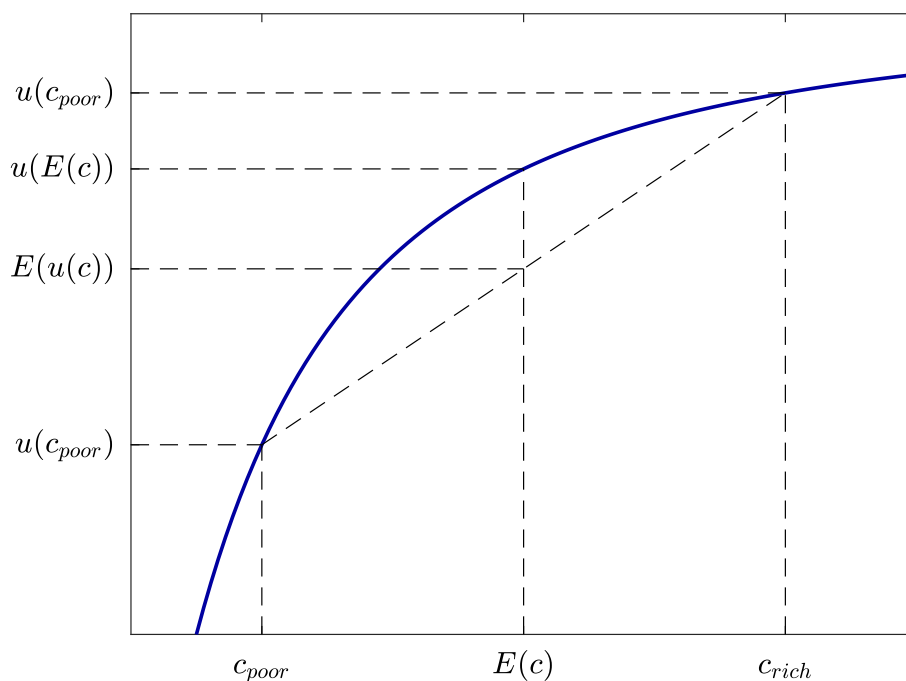
$$u(c, l, a) = \sum_{n=1}^N \frac{1}{N} \left(\bar{u} + \frac{c_n^{1-\sigma}}{1-\sigma} - \theta(1-l)^2 \right) a$$

- Concave utility functions (parameter σ): **risk aversion** (inequality in a country)

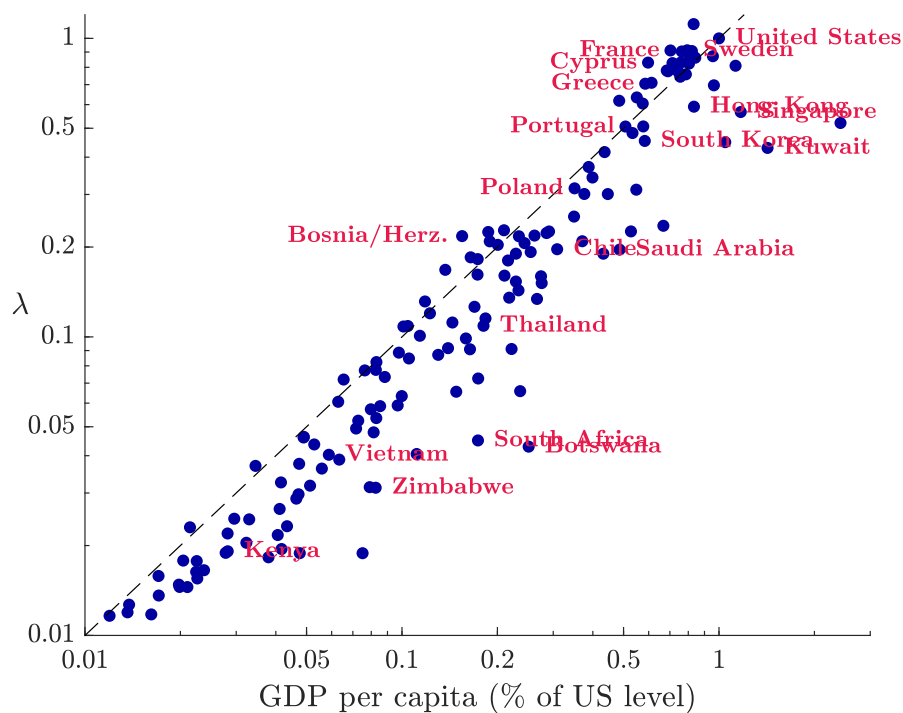
$$u(\mathbb{E}(c)) > \mathbb{E}(u(c))$$

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ ↺ 🔍 ↻

Jones & Klenow's Methodology: Veil of Ignorance, Inequality and Risk Aversion

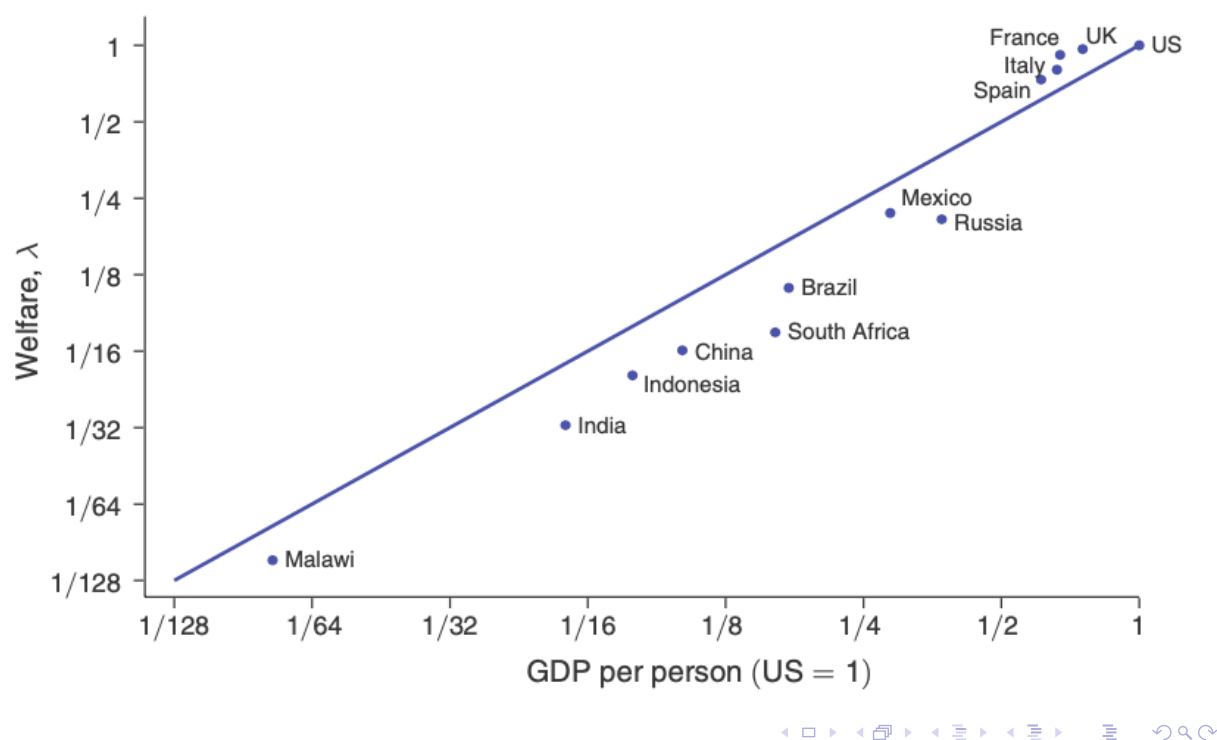


Jones & Klenow's Results



Jones & Klenow's Results

Panel A. Welfare and income are highly correlated at 0.98



Jones & Klenow's Results

Panel B. But this masks substantial variation in the ratio of λ to GDP per capita

